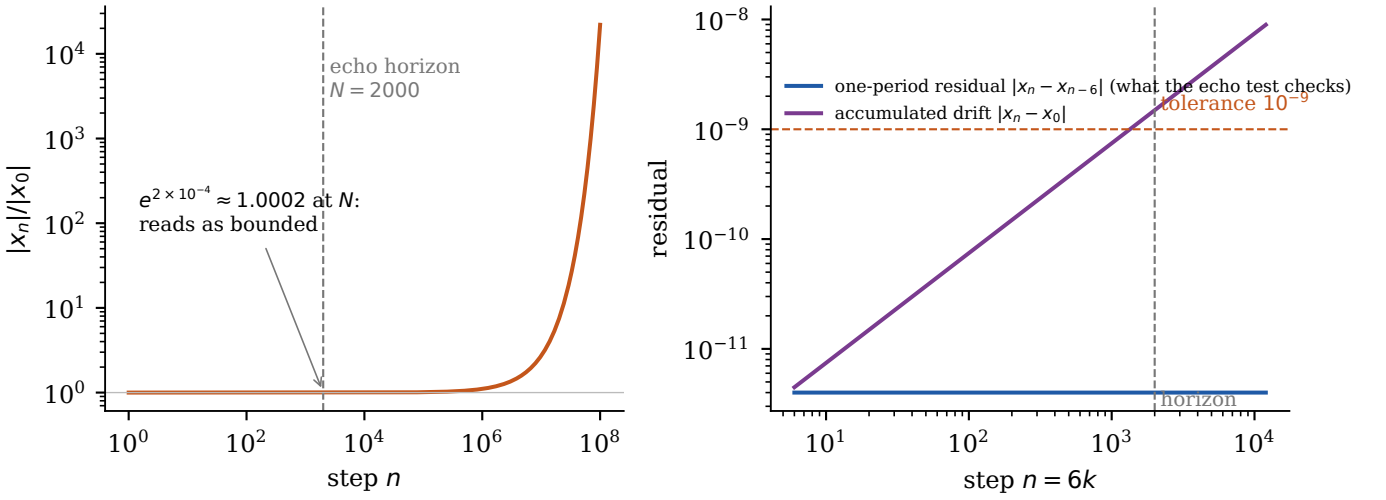
(a) horizon trap, $\rho = 1 + 10^{-7}$: exact class divergent
echo horizon
$N = 2000$
$e^{2 \times 10^{-4}} \approx 1.0002$ at N:
reads as bounded
$|x_n|/|x_0|$
step n
(b) resolution trap, $\tau = 1 + 10^{-12}$: exact class bounded
one-period residual $|x_n - x_{n-6}|$ (what the echo test checks)
accumulated drift $|x_n - x_0|$
tolerance 10^{-9}
horizon
residual
step $n = 6k$